

TASK 1

Question: To register handlers, configure signal actions, handle interrupts, manage signal context, verify stability, and test signal events.

```
ganesh@DESKTOP-79N3772:~$ nano signal.sh
[1]+  Done                  sleep 300
ganesh@DESKTOP-79N3772:~$ chmod +x signal.sh
ganesh@DESKTOP-79N3772:~$ ./signal.sh
Process started. PID: 1034
Press Ctrl+C to send SIGINT.
^CSIGINT received - Signal handled successfully!
```

```
#!/bin/bash

trap 'echo "SIGINT received - Signal handled successfully!"' SIGINT

echo "Process started. PID: $$"
echo "Press Ctrl+C to send SIGINT."

while true
do
    sleep 2
done
```

```
^CSIGINT received - Signal handled successfully!
^Z
[1]+  Stopped                ./signal.sh
ganesh@DESKTOP-79N3772:~$ kill %1
[1]+  Stopped                ./signal.sh
[1]+  Terminated           ./signal.sh
ganesh@DESKTOP-79N3772:~$
```

Observation: The `trap` command registers a handler for `SIGINT` and allows the interrupt signal to be handled without immediately terminating the process.

TASK 2

Question: To capture `SIGINT`, forward signals, protect shell process, terminate foreground jobs, update job states, and test interrupt handling.

```

CSIGINT received - Signal handled successfully!
Z
1]+ Stopped                  ./signal.sh
anesh@DESKTOP-79N3772:~$ kill %1

1]+ Stopped                  ./signal.sh
1]+ Terminated              ./signal.sh
anesh@DESKTOP-79N3772:~$ sleep 300 &
1] 1117
anesh@DESKTOP-79N3772:~$ jobs -l
1]+ 1117 Running              sleep 300 &
anesh@DESKTOP-79N3772:~$ kill -STOP %1

1]+ Stopped                  sleep 300
anesh@DESKTOP-79N3772:~$ jobs
1]+ Stopped                  sleep 300
anesh@DESKTOP-79N3772:~$ kill -CONT %1
anesh@DESKTOP-79N3772:~$ jobs
1]+ Running                  sleep 300 &
anesh@DESKTOP-79N3772:~$ fg %1
sleep 300
C
anesh@DESKTOP-79N3772:~$ jobs
anesh@DESKTOP-79N3772:~$ kill %1 2>/dev/null
anesh@DESKTOP-79N3772:~$

```

Observation: SIGINT interrupts foreground processes, while SIGSTOP and SIGCONT are used to stop and resume jobs and update their execution state.

SESSION 20

TASK 1

Question: To capture SIGTSTP, suspend processes, update job table, preserve state, resume later, and test suspension workflow.

```

ganesh@DESKTOP-79N3772:~$ sleep 300
^Z
[1]+  Stopped                  sleep 300
ganesh@DESKTOP-79N3772:~$ jobs -l
[1]+  1203 Stopped              sleep 300
ganesh@DESKTOP-79N3772:~$ bg %1
[1]+  sleep 300 &
ganesh@DESKTOP-79N3772:~$ jobs
[1]+  Running                  sleep 300 &
ganesh@DESKTOP-79N3772:~$ fg %1
sleep 300
^Z
[1]+  Stopped                  sleep 300
ganesh@DESKTOP-79N3772:~$ jobs -l
[1]+  1203 Stopped              sleep 300
ganesh@DESKTOP-79N3772:~$ bg %1
[1]+  sleep 300 &
ganesh@DESKTOP-79N3772:~$ jobs
[1]+  Running                  sleep 300 &
ganesh@DESKTOP-79N3772:~$ kill %1
[1]+  Terminated              sleep 300
ganesh@DESKTOP-79N3772:~$

```

TASK 2

Question: To create process groups, assign group IDs, transfer terminal control, restore ownership, coordinate signals, and validate behavior.

```

ganesh@DESKTOP-79N3772:~$ kill %1
[1]+  Terminated              sleep 300
ganesh@DESKTOP-79N3772:~$ sleep 300 &
[1] 1294
ganesh@DESKTOP-79N3772:~$ ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
  PID   PPID   PGID   SID   TPGID  STAT  CMD
   332    331    332    332    1304  Ss    -bash
  1294    332   1294    332    1304  S     sleep 300
  1304    332   1304    332    1304  R+    ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
ganesh@DESKTOP-79N3772:~$ sleep 300 &
[2] 1314
ganesh@DESKTOP-79N3772:~$ ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
  PID   PPID   PGID   SID   TPGID  STAT  CMD
   332    331    332    332    1326  Ss    -bash
  1294    332   1294    332    1326  S     sleep 300
  1314    332   1314    332    1326  S     sleep 300
  1326    332   1326    332    1326  R+    ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
ganesh@DESKTOP-79N3772:~$ kill -STOP %1

[1]+  Stopped                  sleep 300
ganesh@DESKTOP-79N3772:~$ jobs -l
[1]+  1294 Stopped (signal)     sleep 300
[2]-  1314 Running              sleep 300 &
ganesh@DESKTOP-79N3772:~$ kill %1 %2

[1]+  Stopped                  sleep 300
[1]+  Terminated              sleep 300
[2]-  Terminated              sleep 300
ganesh@DESKTOP-79N3772:~$ jobs
ganesh@DESKTOP-79N3772:~$

```

Observation: Process groups allow related processes to be controlled together, while `ps` displays their PID, PGID, SID and terminal-related information.